

Civic Issue Reporting System

Empowering Communities Through Digital Governance

Synopsis submitted in partial fulfillment of the requirement for the
Minor Project, VIIth Semester



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1. Introduction

Keeping a city running smoothly is not an easy task. Municipal authorities are responsible for maintaining essential public infrastructure such as roads, streetlights, drainage systems, sanitation facilities, and other civic services. However, in many cities and towns across India, complaint management still relies on traditional methods such as paper records, phone calls, or manual registers. These processes are often slow, difficult to track, and lack transparency, causing common civic issues like potholes, overflowing garbage bins, broken streetlights, and damaged public property to remain unresolved for long periods.

To address this challenge, we are developing the **Civic Issue Reporting System**, a full-stack web application that digitizes the entire process of reporting and resolving civic complaints. The platform aims to create a direct and efficient communication channel between citizens and municipal authorities.

Using the system, citizens can report issues within seconds by simply capturing a photo, allowing GPS-based location tagging, selecting the relevant issue category, and providing a short text or voice description. Once submitted, the complaint is automatically recorded and routed to the appropriate department.

On the administrative side, municipal staff and department heads can access a centralized dashboard that provides a live map of reported issues, advanced filtering options, automated department-wise assignment, and real-time status tracking. This helps authorities respond more quickly while ensuring greater transparency and accountability throughout the resolution process.

This project is being developed as part of **Smart India Hackathon (SIH) 2025** under **Problem Statement SIH25031**, proposed by the **Government of Jharkhand** under the theme "**Clean & Green Technology**." The solution aligns with national initiatives such as **Digital India** and the **Smart Cities Mission**, while also contributing to **United Nations Sustainable Development Goal 11 (Sustainable Cities and Communities)** by encouraging citizen participation and improving urban governance.

Our team of four final-year Computer Science and Engineering students is developing this project as part of the **VII Semester Minor Project**. The application is being built using the **MERN technology stack (MongoDB, Express.js, React.js, and Node.js)** along with **Next.js** for enhanced performance and server-side rendering. The final outcome will be a fully functional civic issue reporting platform, an administrative dashboard, and comprehensive technical documentation.

2. NEED OF THE PROJECT

The need for a dedicated **Civic Issue Reporting System** arises from the challenges citizens and municipal authorities face with existing complaint management processes. In many cities, civic complaints are still handled through paper records, phone calls, or manual registers, making the process slow, inefficient, and difficult to track. As a result, common issues such as potholes, broken streetlights, drainage problems, and uncollected garbage often remain unresolved for long periods. Citizens frequently struggle to know whether their complaints have been received or acted upon, while municipal departments lack a centralized system to monitor and manage issues effectively. A digital platform that enables quick reporting, transparent tracking, and efficient resolution of civic problems can help bridge this gap, improve accountability, and enhance the overall quality of urban services.

2.1 Critical Civic Gap

Citizens encounter civic issues such as potholes, broken streetlights, overflowing garbage bins, and damaged public infrastructure on a regular basis. However, existing complaint-reporting mechanisms are often inefficient, fragmented, or unavailable, particularly in smaller towns and municipalities. Most local authorities still depend on traditional methods such as paper records, phone calls, and in-person visits to register complaints. These manual processes make it difficult to track issues effectively, delay response times, and limit the ability of municipal staff to prioritize problems based on urgency and location. As a result, many civic issues remain unresolved for extended periods, negatively affecting public safety and the overall quality of life for residents.

2.2 Government Priority

This project directly supports major national initiatives like Digital India and the Smart Cities Mission. By launching Smart India Hackathon Problem Statement SIH25031, the Government of Jharkhand is actively looking for tech-driven, crowdsourced tools to boost public involvement and administrative openness, showing a clear, real-world need for this exact solution.

2.3 Real-World Impact

A working civic reporting platform can help millions of city residents by directly connecting people with their local government. It creates clear transparency and accountability by letting everyone track reported problems, assigned teams, and live fix updates, which helps rebuild public trust in local councils.

2.4 Technical Feasibility

Our proposed fix uses proven, open-source web tech—like React, Node, and MongoDB—that our team already knows well. This keeps the project realistic to build and test within our Minor Project timeline, without needing expensive or proprietary software.

3. Problem Statement

Design and build a mobile and web platform that lets residents easily report everyday city issues—like potholes, dead streetlights, trash piles, broken drainage, and damaged public property—using quick photos and automatic GPS tagging. The app needs to auto-route reports to the right council departments, give admins a central dashboard to track and filter active complaints, and send citizens live status updates to make fixes faster, clearer, and far more accountable.

Attribute	Details
Hackathon / Source	Smart India Hackathon (SIH) 2025
Problem Statement ID	SIH25031
Theme	Clean & Green Technology
Category	Software
Proposed By	Government of Jharkhand
SDG Alignment	SDG 11: Sustainable Cities — Target 11.3 (Participatory Urban Planning)

4. Literature Review / Existing Work Analysis

Before finalizing our system design, we reviewed fifteen existing Indian civic-grievance and e-governance platforms—across national, state, local, and private initiatives—to spot common patterns, adoption trends, and key drawbacks. The comparison below sticks strictly to verifiable, publicly documented facts for each platform.

#	Platform	Launching Authority / Scope	Key Features	Identified Gap / Limitation
1	Delhi – PWD Sewa	Public Works Department, Govt. of NCT of Delhi (relaunched July 2026)	Photo- and location-based complaint registration for roads, potholes, waterlogging, drains, flyovers and foot-over-bridges; toll-free 1908 and WhatsApp chatbot support.	Restricted to PWD-maintained infrastructure only; garbage, streetlights (DISCOM-owned) and sanitation fall outside its scope.
2	India – CPGRAMS	Dept. of Administrative Reforms & Public Grievances (DARPG), Govt. of India	24x7 unified portal linking all central ministries, departments and states; auto-routing, unique registration ID, appeal mechanism, UMANG integration and the 2026 “Samadhan Didi” Bhashini-based AI voice chatbot.	A general-purpose administrative grievance workflow — not built for geo-tagged, photo-first, hyperlocal civic-infrastructure reporting or a live issue map.
3	Delhi – PGMS	Government of NCT of Delhi (vertically linked to CPGRAMS)	24x7 grievance registration; vertical integration with CPGRAMS and horizontal routing to district-level subordinate offices; accepts postal complaints.	Text/administrative-workflow based; no photo-evidence capture, GPS tagging, or live map view for citizens.
4	Swachhata–MoHUA App	Ministry of Housing & Urban Affairs with Janaagraha (IChangeMyCity); live in 4,000+ towns	Mandatory live photo (no gallery upload) with auto location tag for sanitation issues; category-wise SLAs (e.g., 12 hrs for garbage); voting, feedback/reopen window; feeds Swachh Survekshan rankings.	Scope limited strictly to sanitation/cleanliness categories; does not cover roads, streetlights or drainage infrastructure.
5	NDMC-311	New Delhi Municipal Council (NDMC Smart City Ltd.)	Complaint registration bundled with utility-bill payment (water, electricity, property tax), an emergency helpline directory and traffic/parking information.	Jurisdiction limited to the NDMC area (Lutyens' Delhi); bundling many unrelated services increases complexity for a citizen who wants to report a single issue quickly.
6	UMANG	MeitY & National e-Governance Division, Govt. of India	Single super-app offering 2,200+ central/state services (2026 update); Aadhaar-based authentication; multilingual “AI-Sathi” voice assistant (Bhashini, 22 languages); CPGRAMS grievance filing built in.	Extremely broad, multi-department scope leads to a complex setup and, per user reviews, frequent slowness/instability for a citizen wanting to report one civic issue.
7	Smart City 311 (Open311 model)	Various Urban Local Bodies under the Smart Cities Mission — e.g., Indore 311, Ahmedabad 311, on the Civic-Solutions Open311-compliant platform	Photo + GPS complaint submission, upvoting/following others' complaints, nearby-service locator, SLA-based routing to the zonal/ward officer.	Fragmented — a separate branded app per city with no shared national data standard, and recurring stability complaints from users.
8	MyNeta (Civic Engagement App)	Citizen First Technologies Pvt. Ltd. — a private civic-tech venture, distinct from the	One-click photo/video complaint capture with geo-tagging; community upvoting; a public accountability “score” for local representatives and	A very recent (2026), privately run platform not yet integrated with official municipal back-ends;

#	Platform	Launching Authority / Scope	Key Features	Identified Gap / Limitation
		unrelated ADR “MyNeta.info” candidate-affidavit site	officials based on issues reported and resolved.	effectiveness depends on voluntary participation of elected representatives.
9	MCD-311	Municipal Corporation of Delhi	Reports illegal dumping, municipal solid waste, potholes, streetlights and stray cattle; multi-agency monitoring by zonal Deputy Commissioners; mandatory after-photo upload by officials; Open311-based.	User reviews report frequent app crashes and complaints closed without genuine resolution; the citizen has no control over complaint closure.
10	e-NIVARAN	Income Tax Department / Central Board of Direct Taxes, Govt. of India (launched 2016)	Unified single-window grievance system for taxpayer complaints (refunds, e-filing, PAN/TAN, TDS) with automated routing and a 30-day resolution target.	Domain-specific to income-tax grievances; included here only as a mature single-window e-governance model — it has no civic-infrastructure reporting scope at all.
11	Karnataka Janaspandana (iPGRS)	Centre for e-Governance, Govt. of Karnataka (launched 1 Nov 2021)	Covers 40+ department schemes including municipal services (water, drainage, garbage, streetlights); integrated with CPGRAMS; SAKALA Act-based service guarantee with appellate escalation.	Routes grievances to HoDs/Secretaries through an administrative workflow; no citizen-facing live map or automatic photo/GPS-based issue categorisation.
12	Odisha – e-Abhijoga	Chief Minister's Office, coordinated by the PG & PA Department and CMGI, Govt. of Odisha	Anytime/anywhere grievance lodging and status tracking; administrative module for the CM office and coordination offices; citizen feedback/rating module.	Broad scheme/service-grievance design without a dedicated photo- or geotag-based workflow for municipal civic-infrastructure complaints.
13	Gujarat – SWAGAT / Municipal i-Grievance Systems	SWAGAT: Chief Minister's Office, Govt. of Gujarat (since 2003); civic complaints also handled separately by AMC, SMC, VMC and RMC portals	SWAGAT connects the CM with district/sub-district officers via monthly video-conferenced grievance hearings; municipal portals auto-assign civic complaints (water, drainage, streetlights, garbage) by ward and category.	No single statewide civic-issue app — citizens must know whether to use the state SWAGAT portal or their own municipal corporation's separate system.
14	Bengaluru – BBMP Sahaaya 2.0 (Namma Bengaluru)	BBMP-IT, Bruhat Bengaluru Mahanagara Palike	Unified app integrating grievances of BBMP, BESCOM, BWSSB, BMTC, BMRCL, BMRDA and BDA; text/photo/video complaints; day-to-day status tracking.	Preceded by overlapping apps for the same city (Fix My Street, PACE, Raste Gundi Gamana) that caused citizen confusion; earlier versions faced server and login issues.
15	Maharashtra – Aaple Sarkar Grievance Redressal Portal	Chief Minister's Office, Govt. of Maharashtra — part of the wider Aaple Sarkar e-governance portal (770+ services)	Token-based grievance filing and tracking; 21-working-day resolution target; satisfaction feedback with escalation on dissatisfaction; mobile app available; covers civic categories (roads, drainage, water) alongside scheme grievances.	A general, text-based state-grievance workflow rather than a dedicated photo/GPS-first civic-reporting tool with a live issue map.

4.1 Gap Analysis

The reviewed platforms fit into four main buckets. First, national and state grievance systems—like CPGRAMS, Delhi PGMS, Karnataka's Janaspandana, Odisha's e-Abhijoga, Gujarat's SWAGAT, and Maharashtra's Aaple Sarkar—focus on broad, text-based workflows across all government offices rather than local, photo- and GPS-led civic reporting with live mapping. Second, niche apps like Swachhata-MoHUA (sanitation only), PWD Sewa (roadways and infrastructure only), and e-NIVARAN (tax complaints, which have no civic link) cover a very narrow scope. Third, city-specific "311" platforms—such as MCD-311, NDMC-311, the Indore/Ahmedabad 311 tools, and BBMP Sahaaya 2.0—are closer to what we are building, but they stay isolated by city, lack a unified system, and often face performance bugs, duplicate apps, and complaints marked closed without real fixes. Fourth, UMANG acts as a broad multi-service portal, while MyNeta is a private civic app that lacks direct integration with local councils and relies purely on voluntary responses from politicians.

The proposed Civic Issue Reporting System tackles these gaps using several key features:

- **All-in-One Category Coverage:** Unlike Swachhata (sanitation only) or PWD Sewa (roads only), a single platform handles roads, streetlights, drainage, waste, and public property together.
- **Budget-Friendly:** Built using completely open-source tools, removing high setup costs and recurring software licensing fees.
- **Fully Customizable:** Complete code access lets any local council adapt the tool to fit its own workflow, ending the endless mix of separate city apps like MCD-311 or NDMC-311.
- **Quick and Easy Setup:** Simple deployment with minimal tech hassle makes it practical for smaller towns alongside major smart cities.
- **Smart Auto-Routing with Live Mapping:** Automated complaint routing to the right department paired with a live interactive map—a powerful duo missing from text-heavy government portals.

5. Proposed System & Methodology

The proposed system connects two main tools—a citizen reporting interface and a municipal admin dashboard—using a shared backend and an automated routing engine, as outlined below.

5.1 Citizen Reporting Interface

- **Camera Capture:** Snap a clear photo directly in the app to show visual proof of the problem.
- **Auto GPS Tagging:** Automatically grabs precise location coordinates so crews know where to go.
- **Voice and Text Notes:** Add extra details by writing a quick message or recording a short voice clip.
- **Simple Category Dropdown:** Pick the issue type—like roads, streetlights, trash, or drainage—from an easy list.

5.2 Municipal Administrative Dashboard

- **Interactive Live Map:** Displays reported issue pins across the city alongside heatmaps highlighting high-priority spots.
- **Smart Filtering Tools:** Filter active reports quickly by category, neighborhood, urgency, current status, or date range.
- **Automated Issue Routing:** Smart system assignment automatically sends incoming complaints to the correct department team.
- **Live Status Alerts:** Instant notification updates sent directly when task assignments or repair progress changes.

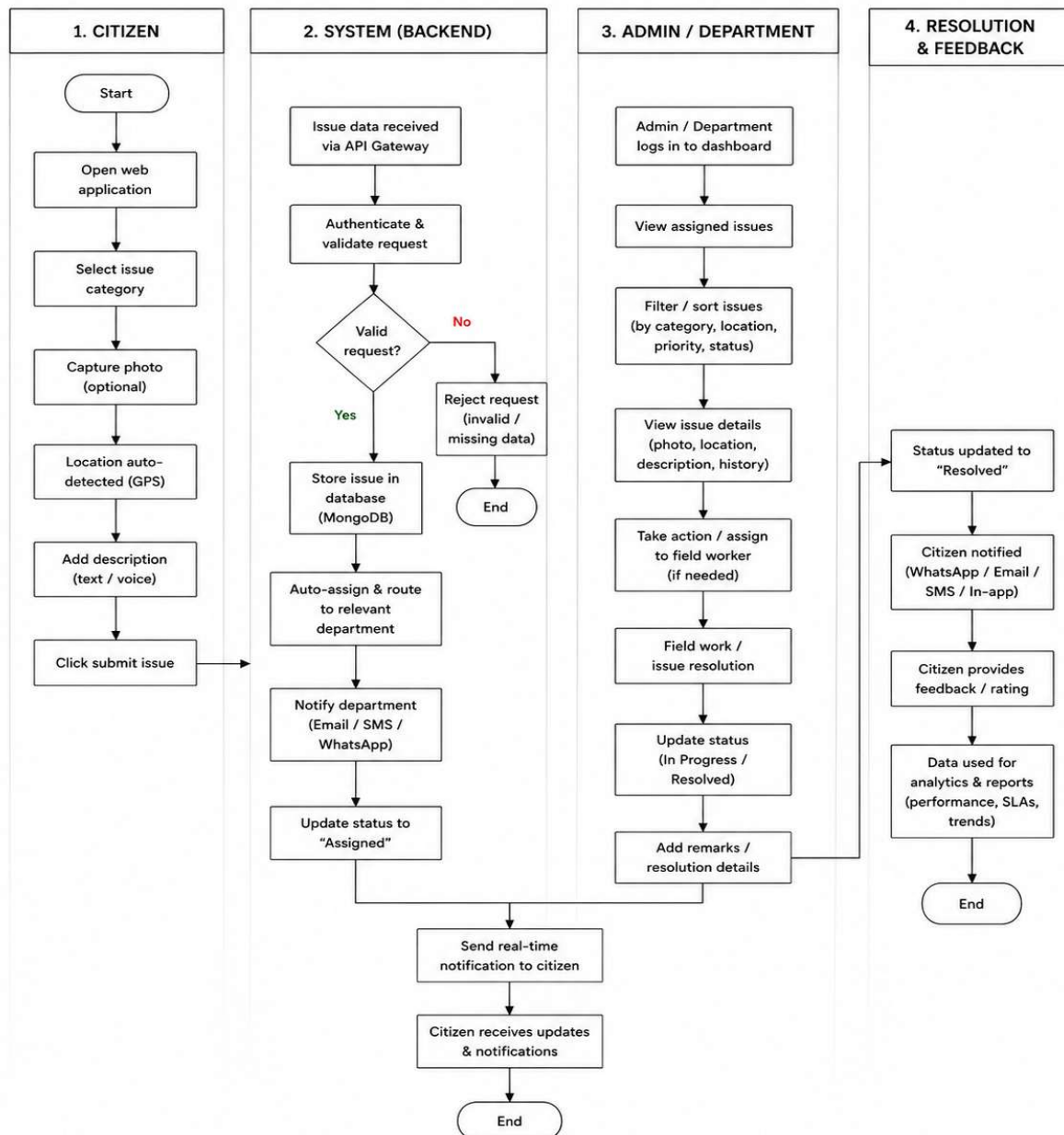
5.3 System Workflow

- **1. Citizen Submission:** The resident submits a complaint using the mobile or web app interface.
- **2. Data Validation:** The backend validates, processes, and safely stores the report details.
- **3. Automatic Routing:** An automated engine assigns the issue to the correct municipal department.
- **4. On-Site Resolution:** Field crews receive the task and carry out the physical repair work.
- **5. Citizen Notification:** The resident receives status updates through a three-stage notification setup, backed by real-time tracking and trend analytics.

5.4 Use Cases and Applicability

The system handles a wide variety of civic issues, including municipal services like potholes, streetlights, trash pickup, and drainage, public safety hazards like broken signs and flooding, environmental concerns like illegal dumping and pollution, and general upkeep of roads, parks, and public property. Main user groups include everyday citizens who report problems and track progress, municipal staff who manage and update tasks, department heads who oversee workloads, and city leadership using big-picture data to guide policies and budgets. The platform is designed to be rolled out across major cities under the Smart Cities Mission, local urban municipalities, town councils in smaller communities, and regional urban development authorities.

5.5 Architecture



6. HARDWARE & SOFTWARE REQUIREMENTS

6.1 Hardware Requirements

Component	Specification
Development Machine	Intel Core i5 (or equivalent) processor, 8 GB RAM minimum, 256 GB SSD storage
Server / Deployment	Cloud-hosted virtual instance (minimum 2 vCPU, 4 GB RAM) for hosting backend and database
Client Device (Citizen)	Any smartphone or desktop/laptop with a camera, GPS/location services and an internet connection
Client Device (Admin)	Desktop/laptop with a modern web browser and stable internet connectivity

6.2 Software Requirements

Layer	Technology
Frontend	Next.js (server-side rendering), React.js (admin dashboard), Tailwind CSS (responsive UI styling)
Backend	Node.js (runtime), Express.js (REST API framework)
Database	MongoDB (NoSQL database)
Authentication & Storage	JWT-based authentication, Multer for file uploads, local storage for media
Maps & Location	Google Maps API / Mapbox for location services
AI/ML Component	Auto-categorization and department-routing of reported issues
Communication	REST API for client-server communication
Development Tools	Visual Studio Code, Git & GitHub for version control, Postman for API testing
Deployment	Cloud-based deployment platform (e.g., Vercel/Render/AWS) with CI/CD pipeline

6.3 Justification of Technology Stack

We selected the MERN-based stack because a single JavaScript language across the entire system speeds up building and testing features. Next.js delivers fast page load speeds using built-in routing and server rendering, while MongoDB's flexible design easily stores different complaint types and handles high user growth smoothly. Tools like JWT and Multer handle user logins and image uploads cleanly without relying on expensive cloud services, keeping our costs low. This setup fits our team's existing coding skills perfectly, allowing us to launch within our 16-week timeline using completely free, open-source software.

7. Outcome of the Project

7.1 Product

The primary deliverable of this Minor Project will be a fully functional, ready-to-deploy Civic Issue Reporting System prototype, which includes:

- **Citizen Web Application:** A user-facing platform for reporting city issues complete with photos, GPS coordinates, and category tags.
- **Municipal Administrative Portal:** A dashboard featuring a live issue map, smart filters, automated department routing, and real-time notification alerts.
- **Technical & User Documentation:** Comprehensive guides detailing the system architecture, database schema, API endpoints, and step-by-step setup instructions.
- **Public GitHub Repository:** A version-controlled code repository created at launch to enable easy future adoption and customization by municipal bodies.

7.2 DEVELOPMENT ROADMAP

The project will be developed over a period of 16 weeks and divided into four key phases. The first phase will focus on laying the foundation of the system, including UI/UX design, database planning, and backend API development. During the second phase, the core functionalities of the web application will be implemented, such as user authentication and the issue reporting module. The third phase will concentrate on developing the administrative dashboard, integrating features like automated department routing and real-time notifications. Finally, the fourth phase will involve comprehensive testing, bug fixing, deployment of the application, and preparation of the final technical documentation to ensure a stable and fully functional system.

7.3 Research Paper (Prospective)

Based on testing results and pilot feedback, our team plans to publish a technical paper detailing our design methodology, AI auto-routing system, and comparative analysis of existing platforms like CPGRAMS, Swachhata-MoHUA, MCD-311, NDMC-311, UMANG, and BBMP Sahaaya 2.0. Submitting this work to a student or departmental conference will contribute practical research to the growing field of civic-tech and e-governance in India.**8.**

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